

■ CRIME INTELLIGENCE & ANALYTICAL PLATFORM

KSP Datathon 2026 — Hack2Skill

Track 2: AI-Driven Crime Analytics & Visualization Platform

Prize Pool	■10,00,000
Registration Deadline	July 19, 2026
Grand Finale	In-person, Bengaluru
Host	Karnataka State Police / SCRB

"Transform the Karnataka State Police's fragmented, Excel-based crime records into a real-time, AI-powered Strategic Intelligence Hub — enabling proactive policing, network analysis, and spatiotemporal crime visualization across all 31 districts."

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■ AI-Driven Crime Analytics & Visualization Platform

Karnataka State Police (KSP) — Hackathon Solution Architecture

Deployed 100% on Zoho Catalyst

Executive Summary

A cloud-native, AI-powered **Crime Intelligence & Analytical Platform (CIAP)** that transforms fragmented police records into an integrated strategic intelligence hub. This architecture strictly adheres to the KSP Datathon 2026 rules by deploying exclusively on **Zoho Catalyst**, utilizing Catalyst Serverless, AppSail, Data Store, and QuickML.

1. High-Level Architecture Overview (Zoho Catalyst)

[Architecture Diagram — MERMAID]

2. Layered Architecture Breakdown

Layer 1 — Data Ingestion & ETL

Component	Catalyst Service	Purpose
Batch Scheduling	Catalyst Cron	Triggers nightly ETL jobs
Event Triggers	Catalyst Signals	Triggers parsing when new FIR CSV is uploaded
ETL Logic	Catalyst Serverless	Python functions to clean and normalize Excel data

Layer 2 — Data Storage Platform

Store	Catalyst Service	What it Stores
Relational Database	Catalyst Data Store	Incidents, victims, offenders, lat/lon coords
Object Storage	Catalyst Stratus	Raw Excel files, exported PDF reports
Graph Database	Catalyst AppSail (OCI)	Neo4j container for criminal network analysis
Caching Layer	Catalyst Cache	Frequent dashboard queries

[!NOTE]

Since Catalyst lacks a native Graph DB, **Neo4j** is deployed via a custom Docker container on **Catalyst AppSail**, ensuring compliance with the "Docker image deployment" capability.

Layer 3 — AI / ML Intelligence Engine

Capability	Catalyst Service	Implementation
Hotspot Detection	AppSail (Python)	ST-DBSCAN clustering using custom Python ML libraries.
Predictive Risk Scoring	Catalyst Zia AutoML	Tabular model forecasting crime risk per district.
NLP on FIR Text	Catalyst Zia Services	Text Analytics to extract Persons/Locations from narratives.
LLM Query Assistant	Catalyst QuickML	RAG pipeline allowing natural language querying of crime data.
PDF Reports	Catalyst SmartBrowz	Automated generation of crime hotspot reports.

Layer 4 — Backend Services

Service	Catalyst Service	Responsibility
API Routing	Catalyst API Gateway	Rate limiting, routing to Serverless functions

Service	Catalyst Service	Responsibility
Core Business Logic	Catalyst Serverless	Python-based Advanced I/O functions serving JSON
Auth & Users	Catalyst Authentication	Login, signup, and Role-Based Access Control (RBAC)
Alerts	Catalyst Push Notifications	Web push alerts for anomaly spikes

Layer 5 — Frontend Analytics Platform

Feature	Technology	Hosting
SPA Framework	React 18 + Vite	Catalyst Web Client Hosting
Geospatial Maps	Deck.gl + Mapbox GL JS	(Client-side)
Network Graph	Cytoscape.js	(Client-side)
Charts & Dashboards	Apache ECharts	(Client-side)

3. Data Flow Diagrams

Real-Time FIR Ingestion Flow

[Architecture Diagram — MERMAID]

4. Tech Stack Summary Table (Zoho Catalyst Native)

Category	Technology / Catalyst Service
Frontend Hosting	Catalyst Web Client Hosting
Frontend Framework	React + Vite + Deck.gl + Cytoscape.js
API Gateway	Catalyst API Gateway
Authentication	Catalyst Authentication

Category	Technology / Catalyst Service
Compute / API Logic	Catalyst Serverless (Advanced I/O Functions)
Custom Containers	Catalyst AppSail (Hosting Neo4j & ST-DBSCAN)
Relational DB	Catalyst Data Store
Object Storage	Catalyst Stratus
Caching	Catalyst Cache
LLM / RAG	Catalyst QuickML
NLP	Catalyst Zia Services (Text Analytics)
Report Generation	Catalyst SmartBrowz
Job Scheduling	Catalyst Cron
Event Routing	Catalyst Signals

5. Differentiators vs. Non-Catalyst Submissions

By fully embracing the Zoho Catalyst ecosystem, this architecture guarantees **100% compliance** with hackathon deployment rules while maintaining enterprise-grade capabilities:

- 1. **AppSail for Graph Analytics:** Leveraging AppSail to run Neo4j ensures we don't lose powerful network mapping just because a native graph DB isn't available.
- 2. **Zia + QuickML Integration:** Replacing generic OpenAI calls with native Catalyst QuickML RAG ensures the intelligence layer is deeply integrated with the Catalyst Data Store.
- 3. **SmartBrowz Reporting:** Instantly generating PDF briefs for investigators using a native Catalyst service rather than clunky client-side generation.